Run the project on Windows

Docker Desktop with WSL2 · Sbusiso Shinga · 20 October 2026

The pipeline runs in Linux containers. On Windows, run the commands inside Ubuntu on WSL2, which is the Linux environment Docker Desktop already uses. Power BI Desktop, which is Windows-only, then connects to the warehouse.

Repository: https://github.com/Sbusiso-tech/job-market-intelligence-platform.git

1. Install

1. Install **Docker Desktop** and turn on **Use the WSL 2 based engine**. In Settings → Resources → WSL integration, enable your Ubuntu distro. Give Docker at least 6 GB of RAM.
2. Install **Ubuntu** from the Microsoft Store if you do not have it: `wsl --install -d Ubuntu`
3. Inside Ubuntu: `sudo apt update && sudo apt install -y make git`
4. Confirm Docker works from Ubuntu: `docker compose version`

2. First run

In the Ubuntu terminal, not PowerShell:

```
git clone https://github.com/Sbusiso-tech/job-market-intelligence-platform.git
cd job-market-intelligence-platform
cp .env.example .env
make up
make pipeline
make top-skills
```

The first `make up` builds the app image and can take several minutes. `make pipeline` calls the live job APIs, so it needs a network connection. Keyless sources (Himalayas, Jobicy, Remotive, Arbeitnow) are enough. Add Adzuna or RapidAPI keys to `.env` if you have them.

From Windows, open http://localhost:8000/docs and http://localhost:8081. The ports are published by Docker Desktop.

3. Power BI

1. Install Power BI Desktop.
2. Get data → PostgreSQL database. Server `localhost:5432`, database `jobmarket`. User and password `jobmarket`.
3. Import from schema `marts`: `fct_job_postings`, `dim_skill`, `dim_company`, `bridge_job_skill`, `mart_skill_demand`, `mart_skill_demand_monthly`, `mart_salary_by_skill`, `mart_skill_by_job_family`.
4. Relationships: `fct_job_postings[job_id]` 1→* `bridge_job_skill[job_id]`; `dim_skill[skill_key]` 1→* `bridge_job_skill[skill_key]` (both directions); `dim_company[company_key]` 1→* `fct_job_postings[company_key]`.

Suggested pages: top skills, a monthly trend, salary percentiles, skills by role family, and top companies. Measure ideas are in the repository README.

Partner and CRM companies are fictional. For a market-only report, filter `source` to the API sources.

4. The other services

Command (inside Ubuntu)	What you open on Windows
`make airflow`	http://localhost:8080 · admin / admin
`make cdc`	`curl localhost:8000/jobs/latest` from Ubuntu
`make metabase`	http://localhost:3000 · host `postgres`, database `jobmarket`, schema `marts`
`make down`	stops containers, keeps data

The recruitment CRM is a second PostgreSQL on `localhost:5433` (database `crm`, user `crm`). The partner SFTP server is `localhost:2222` (user `partner`).

5. If Docker is not visible in Ubuntu

Docker Desktop → Settings → Resources → WSL integration → enable Ubuntu, then restart Docker Desktop. `docker info` inside Ubuntu should succeed before you run `make up`.